

Rear Drive Axle and Differential



The rear axle assembly consists of the following:

- Integral-type housing hypoid gear design (center of the pinion set below the centerline of the ring gear)
- Hypoid differential ring gear and pinion consisting of a 9.75-inch ring gear and an overhung drive pinion that is supported by 2 opposed tapered roller bearings
- Pinion bearing preload that is maintained by a drive pinion collapsible spacer on the differential pinion shaft and adjusted by the pinion nut
- Cast center section with 2 steel tube assemblies and a stamped differential housing cover
- Differential housing cover using silicone sealant as a gasket
- Differential pinion shaft that is retained by a threaded differential pinion shaft lock bolt attached to the differential case
- Differential case that is mounted in the rear axle housing between 2 opposing differential bearings that are retained in the rear axle housing by 2 removable bearing caps
- Differential bearing preload and ring gear backlash which are adjusted by differential bearing shims that are located between the differential bearing cups and the rear axle housing

The rear drive axle receives rotational torque from the driveshaft. The direction of the rotation determines movement of the vehicle. The rotation enters the rear drive axle through the drive pinion flange, turning the drive pinion and rotating the ring gear. The ring gear is bolted to the differential case. The differential case houses the pinion shaft upon which the differential pinion gears ride. The pinion gears are in mesh with the differential side gears to which the axle shafts are splined. It is the unique arrangement of the differential that allows the axle shafts to be driven at different speed when needed, such as when driving around corners. A Traction-Lok differential is available that introduces a series of clutch plates and discs behind the differential side gears to deliver as much torque as possible to each axle shaft when the traction at the rear wheels is not equal. The Traction-Lok differential requires a friction modifier that must be installed with the axle lubricant for the plates and discs to be functional. An Electronic Locking Differential (ELD) is also available which electronically locks the differential side gears to the differential case which will allow maximum torque to both rear wheels. The operator controls the differential with a dash mounted switch.

The **ELD** system consists of the following components:

- **ELD** coil, located at the rear differential
- Rear differential
- Mode Select Switch (MSS) (unique for **ELD** equipped vehicles), located on instrument panel

The **ELD** system, which is optional on Electronic Shift-On-The-Fly (ESOF) equipped vehicles, allows the operator to lock the rear differential side gears when in 4H or 4L mode. **ELD** operation is requested by pulling out on the **MSS** when in 4H or 4L mode.

The operator can switch between 4H mode without **ELD** or 4H mode with **ELD** at speeds up to 30 km/h (19 mph). In 4H, the **ELD** will disengage at speeds above 40 km/h (25 mph) and will automatically reengage at speeds below 30 km/h (19 mph).

To engage or disengage the **ELD** while in 4L range, the vehicle speed must be less than 5 km/h (3 mph).